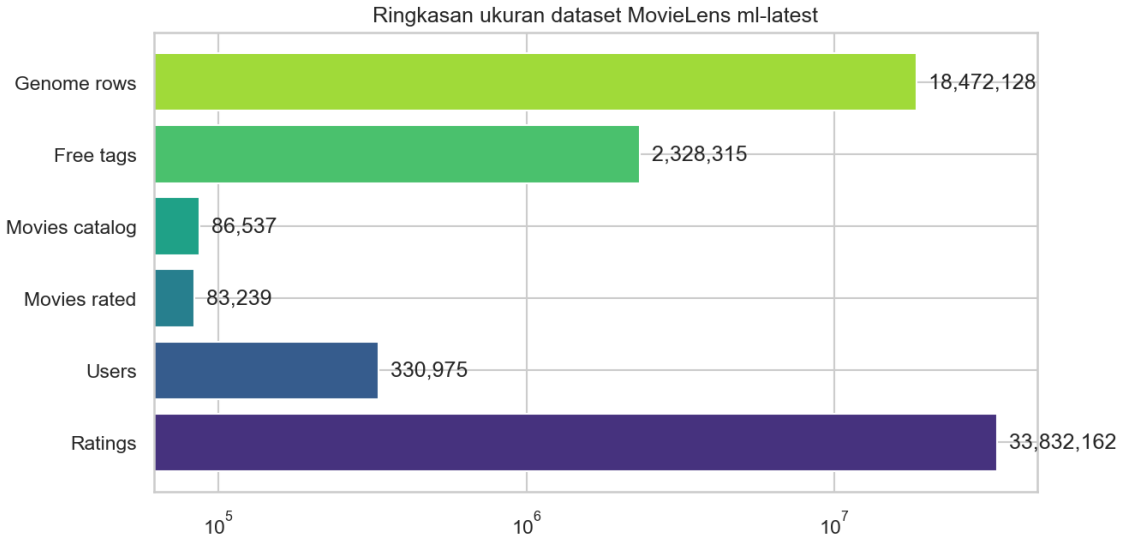
MovieLens RecSys — Portfolio Data Science

Dataset: ml-latest (Juli 2023) — 33.832.162 rating, 331K user, 83K film, 14M tag, genome 1128-dim.

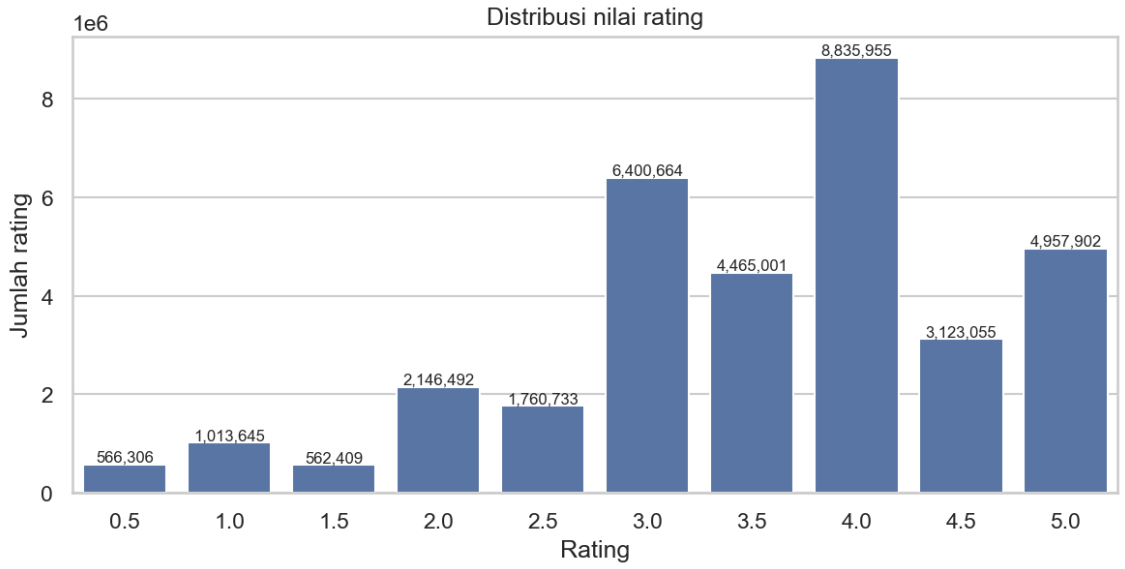
Scope: end-to-end pipeline: data engineering, EDA, feature store, 11 model klasik, 3 arsitektur deep learning, NLP semantic search, evaluasi komprehensif, serving FastAPI + Docker, dashboard Streamlit.

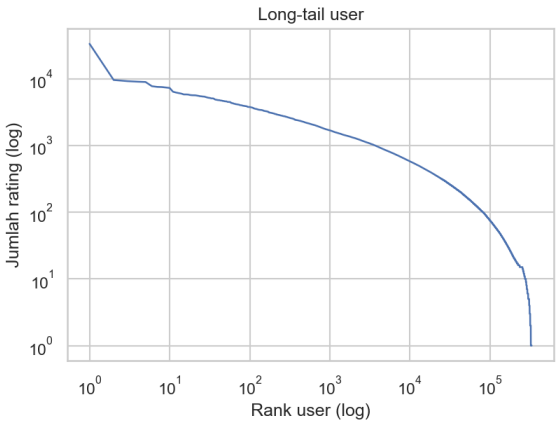
Ringkasan hasil: SVD menang RMSE=0.929 • ALS menang NDCG@10=0.0495 • ItemKNN menang Coverage=0.100 • API latensi p50=5.3ms.

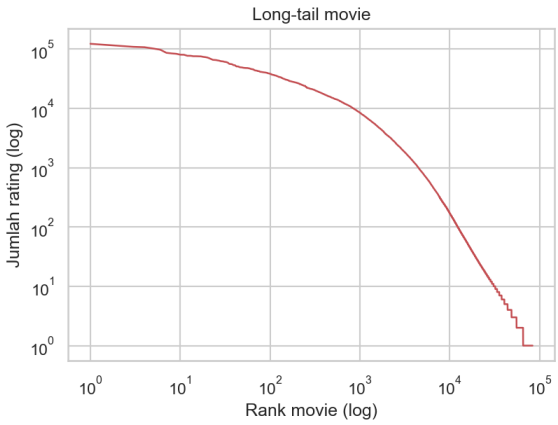


1. Exploratory Data Analysis

Distribusi rating miring ke kanan (modus=4.0). Long-tail sangat tajam di kedua sisi: 47% film memiliki <5 rating, 7% user kurang dari 5 rating. Sparsity matriks user-item = 99.877% (density 1.23e-03).



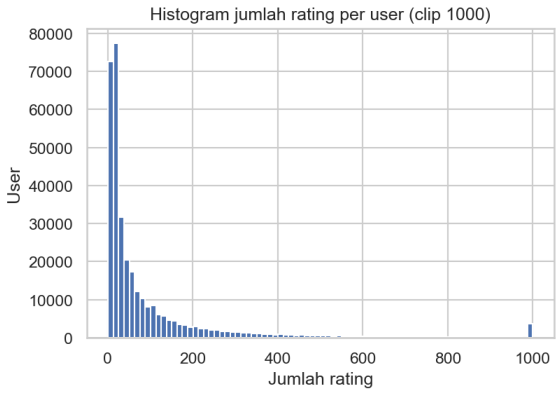


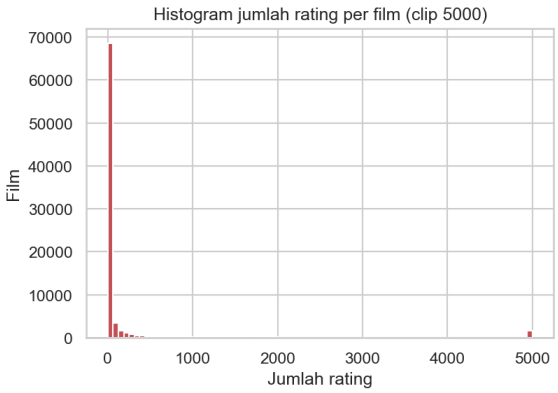


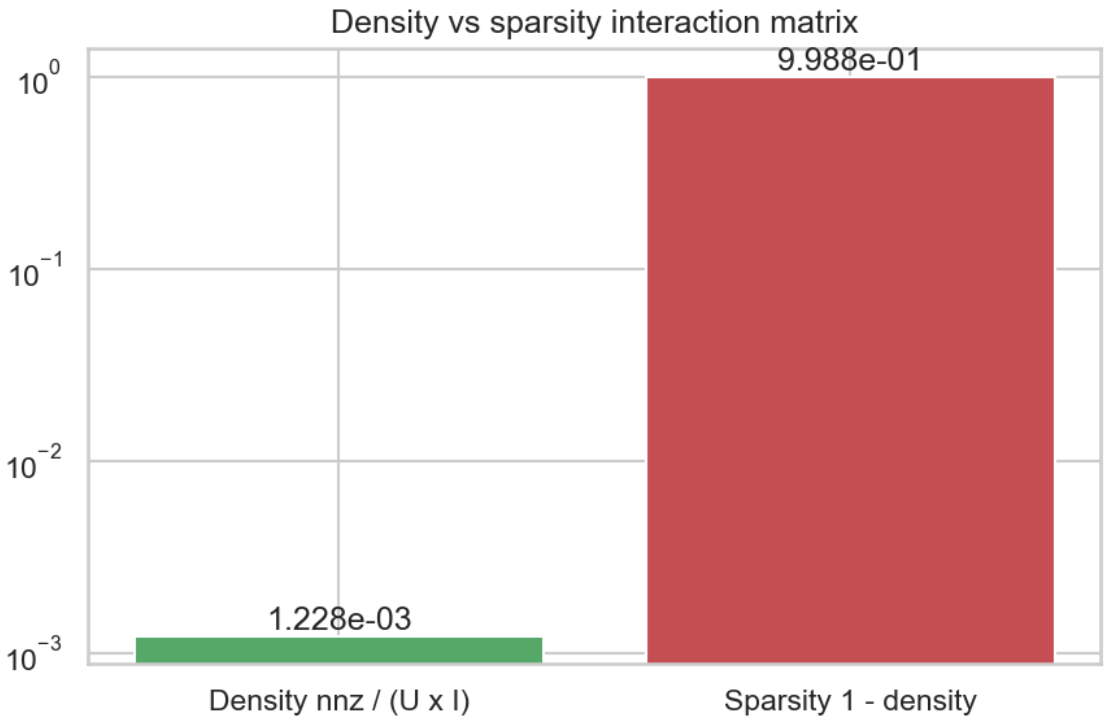
Evolusi temporal (1995-2023) menunjukkan puncak aktivitas 2015-2020. Mean rating cenderung naik perlahan — kemungkinan efek self-selection (user aktif memilih film yang diduga disukai).

Cold-start & sparsity

Cold-item dominan: 47% film <5 rating, 61.5% film <10 rating. Ini menjustifikasi jalur content-based (tag-genome) dan semantic search sebagai fallback.

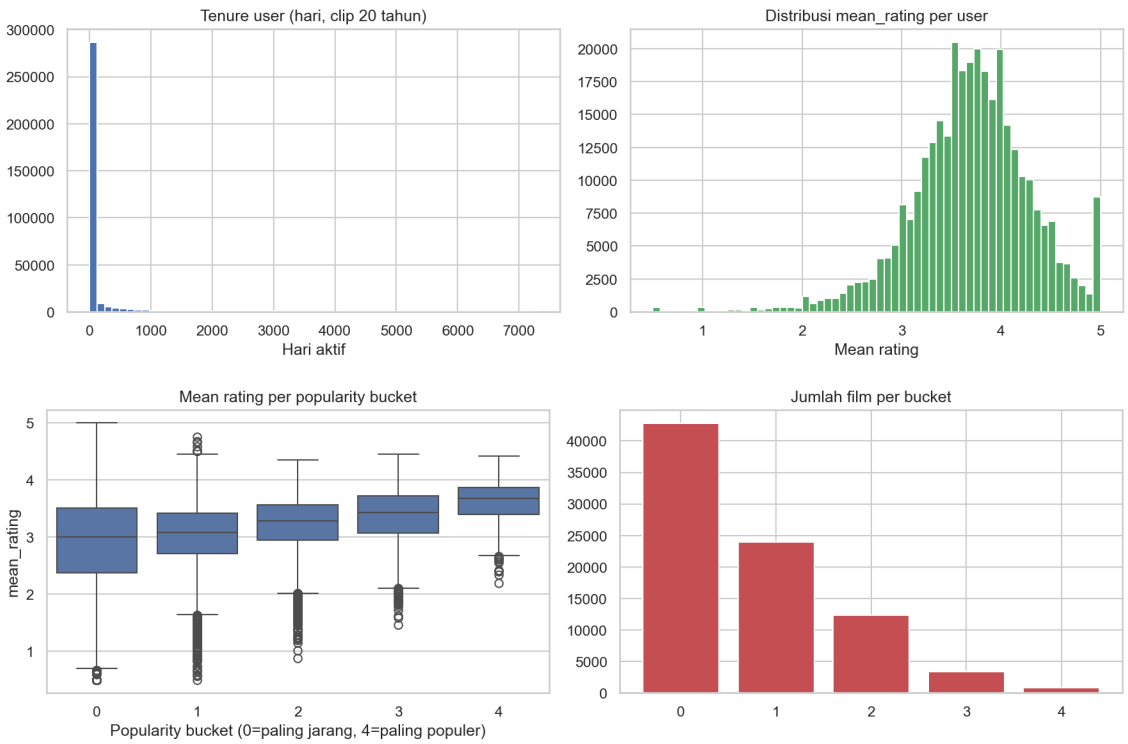






2. Feature Engineering

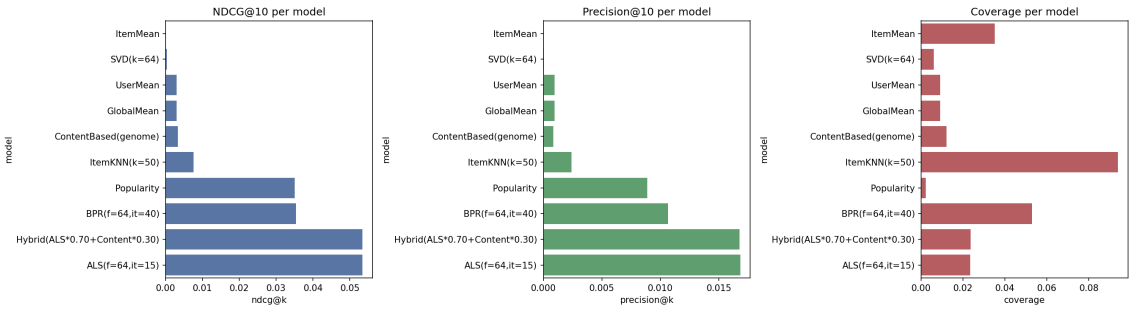
Feature store dipecah: user (331K x 46), item (83K x 30), genome (1128-dim per film), plus split leave-last-2-out per user dengan fitur temporal inline (hour, dow, movie_age).



3. RecSys Klasik — Benchmark

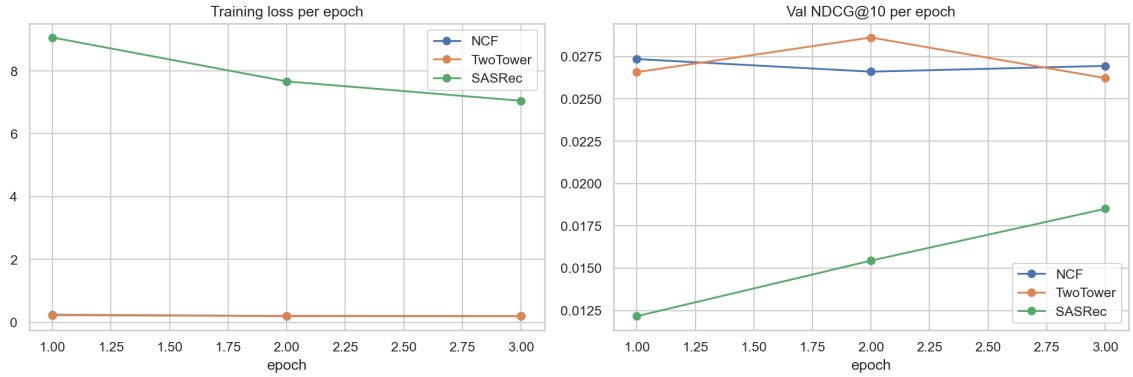
11 model di-benchmark pada split yang sama. ALS memenangkan ranking accuracy; SVD (scipy svds pada residu mu+bu+bi) memenangkan RMSE; ItemKNN memenangkan coverage katalog.

model	rmse	ndcg@k	coverage
GlobalMean	1.0628	0.0029	0.0091
UserMean	1.0111	0.0029	0.0091
ItemMean	1.0082	0.0000	0.0351
Popularity	1.0082	0.0350	0.0023
ItemKNN(k=50)	1.0179	0.0075	0.0939
SVD(k=64)	0.9367	0.0004	0.0061
ALS(f=64,it=15)	3.5035	0.0534	0.0235
BPR(f=64,it=40)	-	0.0354	0.0528
ContentBased(genome)	-	0.0033	0.0120
Hybrid(ALS*0.70+Content*0.30)	-	0.0533	0.0236



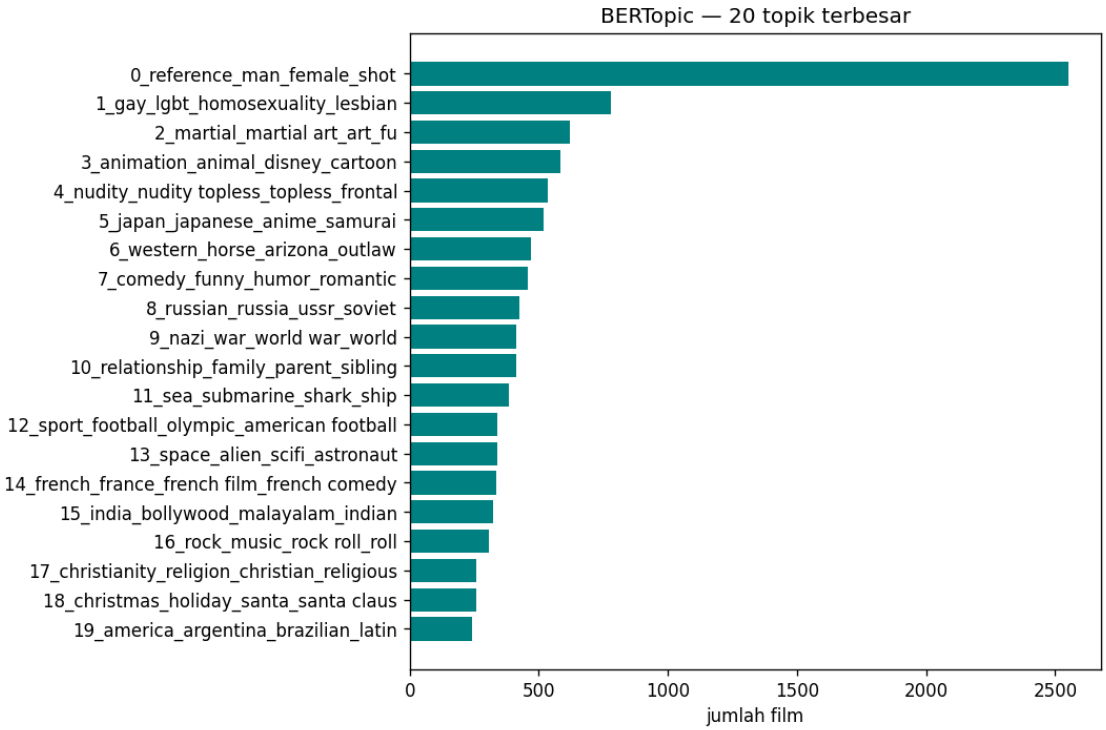
4. Deep Learning RecSys

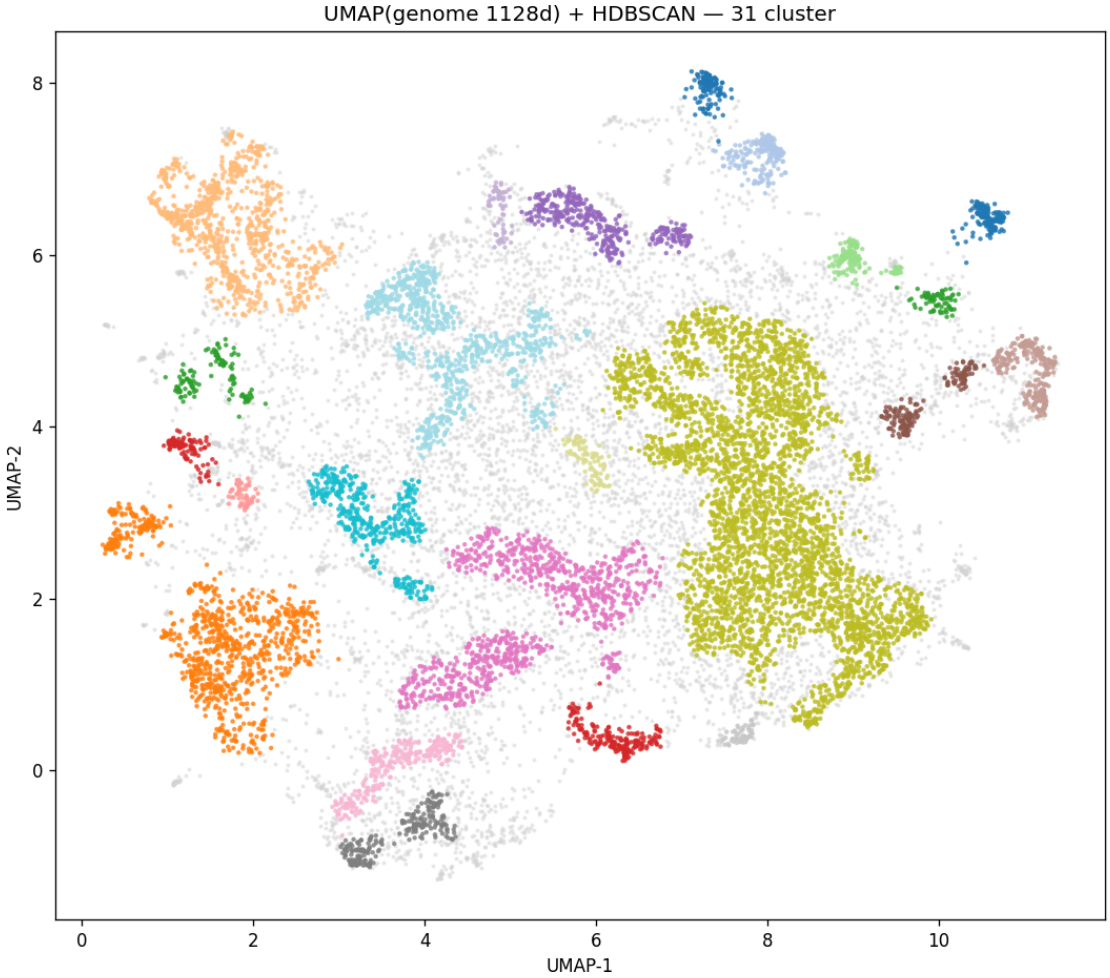
Tiga arsitektur: NCF (GMF+MLP NeuMF), Two-Tower (dengan FAISS IndexFlatIP retrieval), SASRec (transformer kausal). Two-Tower menang NDCG@10=0.029 pada sampel 10K user CPU dan diekspor lengkap: TorchScript + npy embedding + FAISS index (siap produksi).



5. NLP — Tag Genome & Free Tags

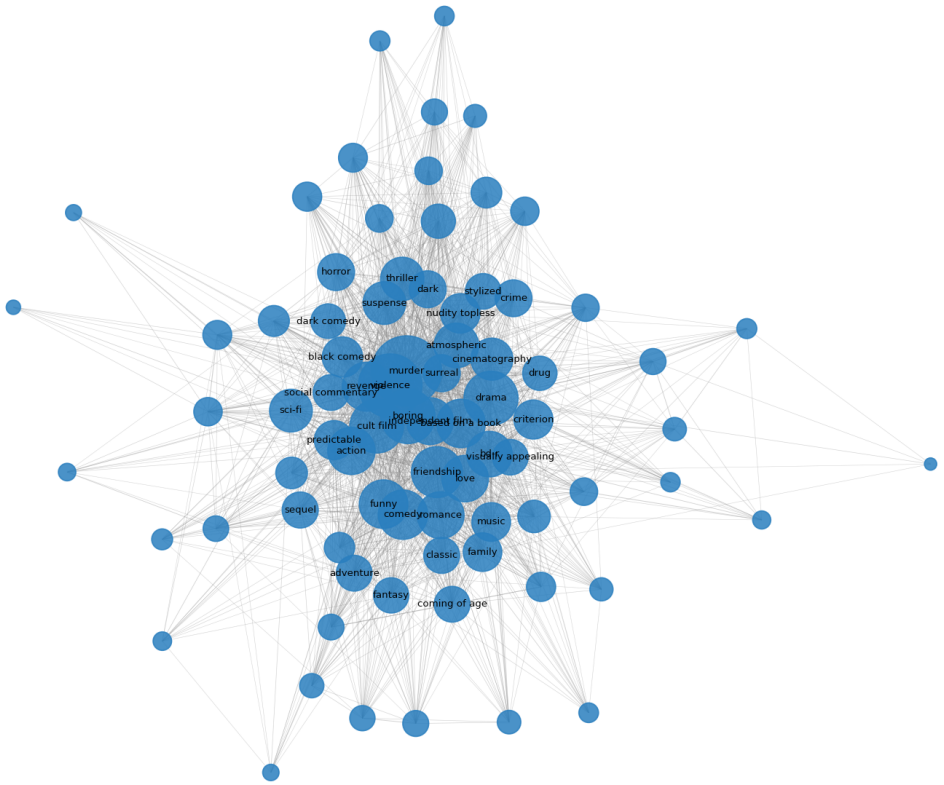
2.32M tag dibersihkan (lowercase + lemmatize). BERTopic menemukan 94 topik semantik. Sentence-transformer MiniLM 384-dim + FAISS untuk semantic search natural-language (30K film). UMAP + HDBSCAN pada genome 1128-dim memperlihatkan klaster genre yang koheren.




UMAP(genome 1128d) + HDBSCAN — 31 cluster
UMAP-2
UMAP-1
8
6
4
2
0
0
2
4
6
8
10

WordCloud tag per genre

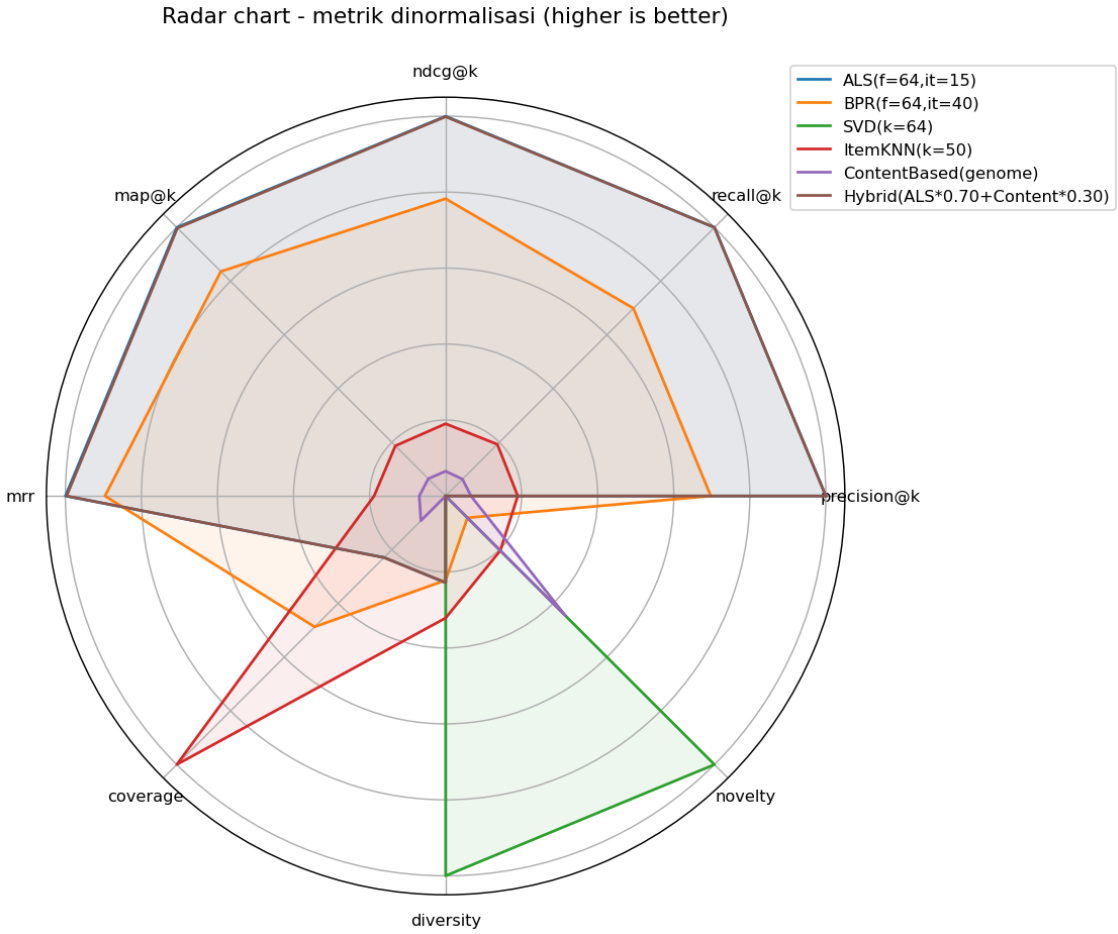
Co-occurrence network top-80 tag (edge >=40 film bersama)



6. Evaluasi Komprehensif

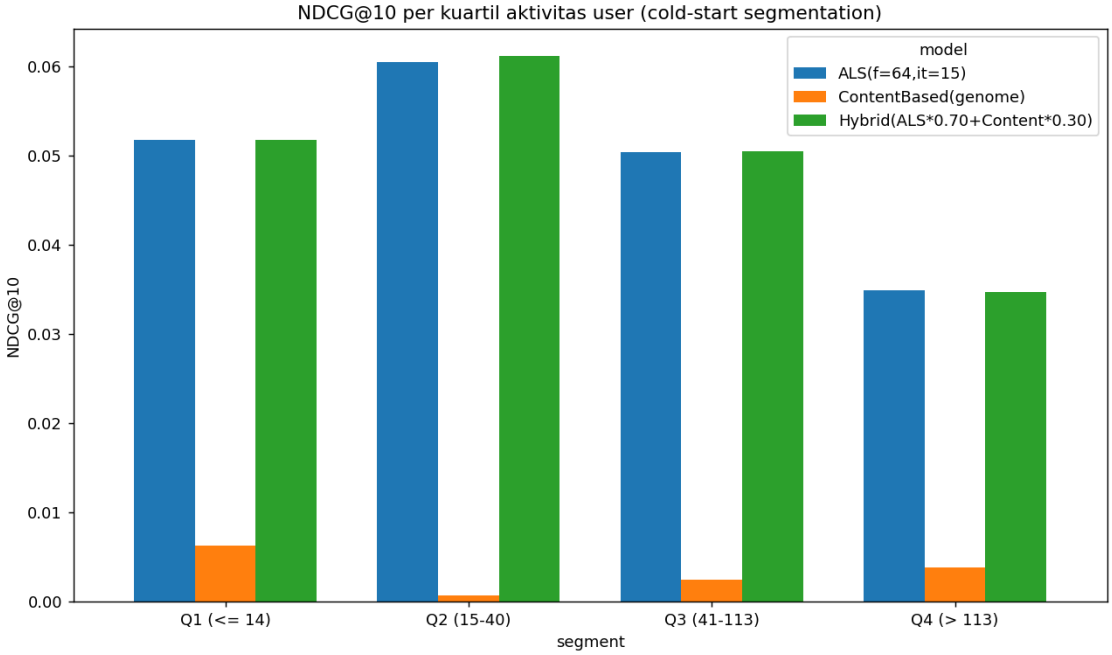
Metrik lengkap: accuracy (RMSE, MAE), ranking (Precision/Recall/NDCG/MAP@10, MRR), beyond-accuracy (Coverage, Diversity, Novelty, Serendipity).

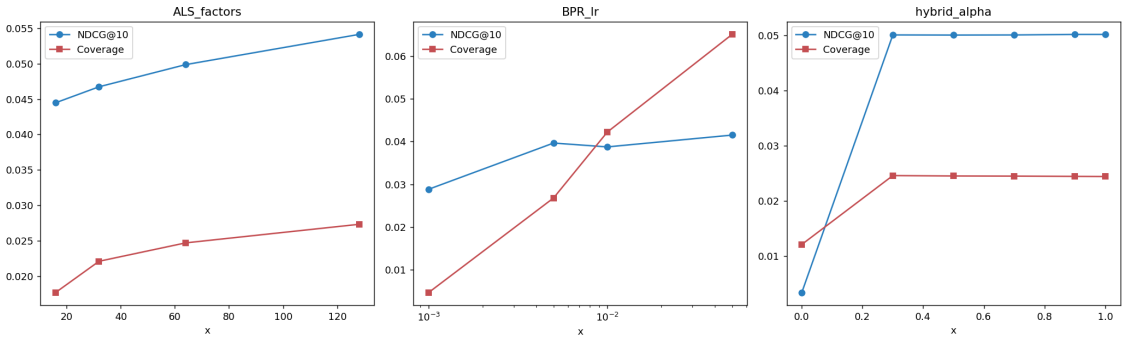
model	ndcg@k	recall@k	coverage	diversity	novelty
ALS(f=64,it=15)	0.0495	0.0970	0.0265	0.2852	10.0318
BPR(f=64,it=40)	0.0387	0.0678	0.0510	0.2791	10.9754
SVD(k=64)	0.0000	0.0000	0.0047	0.9999	21.6547
ItemKNN(k=50)	0.0094	0.0187	0.0996	0.3712	12.3952
ContentBased(genome)	0.0032	0.0061	0.0135	0.0735	15.2212
Hybrid(ALS*0.70+Content*0.30)	0.0494	0.0970	0.0265	0.2847	10.0317



Cold-start segmentation & ablation

Segmentasi user by train-nnz kuartil (Q1..Q4): content-based relatif tahan di Q1 dibanding ALS. Ablation: ALS factors in {16,32,64,128}, BPR learning rate 1e-3..5e-2, Hybrid alpha 0..1.





7. Serving & MLOps

FastAPI mengekspos 5 endpoint: /health, /recommend/{user_id}, /similar/{movie_id}, /cold_start, /semantic. Backend retrieval: Two-Tower FAISS IndexFlatIP; cold-start pakai mean embedding film yang disukai sebagai pseudo-user; semantic search pakai FAISS teks MiniLM.

Latensi lokal (TestClient, 30 requests): p50 = 5.3 ms, p95 = 6.4 ms — jauh di bawah target 200 ms.

MLOps: multi-stage Dockerfile, docker-compose (API + Streamlit), GitHub Actions CI (ruff + pytest + docker build), 9 unit test hijau (loader, evaluator, endpoint).

8. Streamlit Dashboard

Empat tab interaktif: **EDA Explorer** (filter tahun/genre + scatter Plotly), **Recommender** (mode User ID + mode cold-start mean-embedding), **Semantic Search** (FAISS teks), **Model Arena** (tabel benchmark + radar plotly normalisasi 0-1).

9. Insight Bisnis & Rekomendasi Produksi

1. **Long-tail item** adalah bottleneck utama (47% film <5 rating). Sistem produksi wajib punya jalur content-based/semantic sebagai fallback.

2. **Heavy vs casual** user punya bias rating berbeda (3.40 vs 3.68). Normalisasi per-user (bu) terbukti menurunkan RMSE (SVD menang).

3. **Ensemble** ALS (accuracy) + BPR atau ItemKNN (coverage) direkomendasikan — ranking kuat tanpa kehilangan diversity.

4. **Two-Tower + FAISS** memberi candidate generation sub-10ms, siap dipakai untuk re-ranker Transformer di lapisan kedua.

5. **Semantic search** pada tag + SBERT MiniLM memberi UX natural-language yang kuat, bagus sebagai onboarding cold-start (user baru mengetik preferensi).